

Anshu Dahal

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EDUCATION

Truman State University

Bachelor of Science in Computer Science and Data Science

Organizations/Awards: President's Scholarship Recipient, President's Honors List,

Advanced Level Education Scholarship

Kirkville, MO

May 2027

GPA: 4.0 / 4.0

PROFESSIONAL EXPERIENCES

Computational Cancer Genomics Summer Program, Washington University in St. Louis

St. Louis, MO

Research Intern

May 2025 – Jul 2025

- Engineered reproducible data pipelines to clean, standardize, validate, and analyze large-scale clinical, demographic, genomic, survival, and SDOH datasets for breast cancer ML research.
- Built maintainable ML data-processing pipelines, statistical visualizations, validation checks, and technical documentation using R, Quarto, Git, and version-controlled workflows.
- Helped research and prototype an LLM-powered RAG application using LlamaIndex, Hugging Face embeddings, metadata filtering, retrieval, reranking, and an OpenAI-compatible model endpoint.
- Built a balanced evaluation harness and benchmarked model behavior with RAGAS metrics, comparing retrieval algorithms, investigating failure modes, and iterating to improve AI quality.
- Used Docker, Linux, and remote servers via SSH for model development, data processing, pipeline testing, performance experiments, and deployment workflows.

Office of Community and Enrichment, Truman State University

Kirkville, MO

Programming Assistant

Jan 2024 – Present

- Maintained oce.truman.edu, updating website content and supporting a reliable, accessible web experience for students and community users
- Implemented and evaluated ideas for using agentic AI to support website maintenance and make recurring content and maintenance workflows more efficient

Skills: Python, C++, SQL, machine learning, deep learning, computer vision, data science, statistics, PyTorch, torchvision, EfficientNet-B3, Grad-CAM, scikit-learn, pandas, Monte Carlo cross-validation, model evaluation and debugging, data validation, feature and data pipelines, R, Java, Kotlin, FastAPI, REST APIs, Linux, Docker, SSH, Git, GitHub, MongoDB Atlas, Mongoose, relational and NoSQL databases, LLMs, Qwen, Gemini API, LlamaIndex, Hugging Face embeddings

PROJECTS

Student Wellness Check-In (In Development)

Kotlin, Jetpack Compose, MVVM, Room, FastAPI, Qwen SLM, Gemini API

- Developed an Android ML/AI prototype using Kotlin, Jetpack Compose, MVVM, Room, and a Python FastAPI backend, with local data storage, visualizations, and safety-aware user flows.
- Implemented a hybrid on-device/server AI workflow that routes validated requests across Qwen, Gemini, local algorithms, and safe fallbacks while returning structured JSON without exposing API keys.
- Analyzed latency, memory use, structured-output reliability, safety, resource-matching accuracy, API failures, and model-quality tradeoffs to improve performance and user experience.

DDI Dermatology AI Fairness Research

Python, PyTorch, EfficientNet-B3, scikit-learn, Grad-CAM, pandas

- Developed a reproducible computer-vision pipeline in Python and PyTorch, fine-tuning EfficientNet-B3 with MGOT fairness regularization and per-skin-type temperature calibration.
- Designed 10-fold Monte Carlo cross-validation and group-wise statistical evaluation across DDI and Fitzpatrick17k, measuring AUC, error rates, calibration error, and weighted harm.
- Improved reported FST V-VI AUC from 0.61 to 0.77 and reduced false-negative rate from 0.88 to 0.42; implemented data validation, checkpointing, Grad-CAM debugging, reporting, and eight figure workflows.

Memory Garden

React, Node.js, MongoDB, Mongoose, OpenAI API, Emotion Recognition AI

- Developed a full-stack AI application with React and Node.js, combining interactive visualization with an OpenAI-powered journaling workflow.
- Integrated AI summarization to process user logs and provide empathetic feedback.
- Designed MongoDB Atlas and Mongoose schemas for persistent NoSQL storage and cloud-hosted application data.
- Designed intuitive UI/UX interactions with React, focusing on accessibility and emotional resonance.

ACTIVITIES

Google Developers Student Club

Lead, Marketing and Communications Chair

Kirkville, MO

Aug 2025- Present, Jan 2024 - Aug 2025